

YEAR 7 MATHEMATICS

## Chapter 5B

# Equations

MIXED PRACTICE WORKSHEETS

# 5B

## How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

### WE DO

The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

## Contents and progress tracker

| Lesson | Topic                         | Page | Date | Done                     |
|--------|-------------------------------|------|------|--------------------------|
| 5.10   | Guess, check and improve      | 3    |      | <input type="checkbox"/> |
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## In this chapter you will

- check whether a value is a solution
- solve equations by guess, check and improve
- solve one-step and two-step equations using inverse operations
- write and solve equations to solve problems

### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule,  $z \rightarrow z^2 + c$ , thousands of times for every point. Every booklet shows a different part of the set: this one is the valley between the main body and the left bulb, where the seahorses start. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

An equation is true for one value of the pronumeral: the solution. Guess, check and improve: try a value, substitute it, see if it is too big or too small, then try again.

Watch your teacher.  
Copy every step.

**E1** Solve in your head:  
 $x + 7 = 12$

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**E2** Is  $m = 4$  a solution of  
 $3m + 2 = 15$ ?

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**E3** Guess, check and improve:  $6p + 5 = 77$

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## Easy

Questions 1–9

**1** Solve in your head:  $2y = 6$

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**2** Solve in your head:  $\frac{x}{5} = 11$

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**3** Is  $x = 8$  a solution of  
 $3x + 1 = 25$ ?

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**4** Solve in your head:  $w + 2 = 11$

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**5** Solve in your head:  $\frac{x}{4} = 12$

---

**6** Is  $n = 6$  a solution of  
 $5n + 8 = 43$ ?

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**7** Solve in your head:  $10b = 70$

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**8** Solve in your head:  $\frac{n}{2} = 2$

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**9** Find  $k^2$  when  $k = 3$ .

from 5.05

## Easy

Questions 10–18

**10** Solve in your head:  $\frac{n}{6} = 3$

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**11** Is  $p = 3$  a solution of  
 $2p + 3 = 9$ ?

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**12** Solve in your head:  $10w = 10$

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**13** Write 35% as a decimal.

from 4.10

**14** Is  $y = 8$  a solution of  
 $2y + 1 = 17$ ? Show the check.

$2 \times 8 + 1 =$  \_\_\_\_\_

yes / no: \_\_\_\_\_

**15** Solve in your head:  $\frac{t}{6} = 12$

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**16** Is  $p = 6$  a solution of  
 $3p + 9 = 27$ ?

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**17**  $bn \div b$

from 5.09

**18** Solve in your head:  $4p = 12$

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## Medium

Questions 19–25

**19** Is  $y = 8$  a solution of  
 $3y + 7 = 28$ ? Show the check.

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**20** Solve  $6y + 16 = 94$ . Your first guess is 11. Is it too big or too small?

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**21**  $49 + 42 + 51 + 16$

from 5.01

**22** Solve by guess and check:  
 $x^2 = 144$ , where  $x$  is positive.

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**23** Is  $w = 2$  a solution of  
 $2w + 5 = 11$ ? Show the check.

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**24** A rectangle is  $7b$  cm long and  
 $5n$  cm wide. Write its area.

from 5.08

**25** Solve  $7m + 10 = 101$ . Your first guess is 10. Is it too big or too small?

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**Medium**

Questions 26–35

- 26 Solve  $6a + 12 = 48$ . Your first guess is 5. Is it too big or too small?

- 27 Solve by guess and check:  $y^2 = 25$ , where  $y$  is positive.

- 28 Write  $4\frac{2}{5}$  as an improper fraction.

from 4.01

- 29 Is  $b = 6$  a solution of  $6b + 5 = 35$ ? Show the check.

- 30 Work out  $5 \times (30 + 6)$  two ways.

from 5.02

- 31 Guess, check and improve:  $8x + 8 = 112$ . Record each guess.

- 32 Solve  $3n + 12 = 42$ . Your first guess is 7. Is it too big or too small?

- 33 Simplify  $12x - 5x$ .

from 5.07

- 34 Solve by guess and check:  $b^2 = 49$ , where  $b$  is positive.

- 35 Find  $4n + 9$  when  $n = -3$ .

from 5.05

**Challenging**

Questions 36–44

- 36 Guess, check and improve:  $7x + 14 = 56$ . Record each guess.

- 37 Two numbers that are next to each other add to 17. Find them by guess, check and improve.

- 38 Jack scored 50% on a test out of 180. How many marks is that?

from 4.11

- 39 A number multiplied by itself, plus the number, is 56. Find the number.

- 40  $9a \div a$

from 5.09

- 41 Guess, check and improve:  $4x + 6 = 42$ . Record each guess.

- 42 Two numbers that are next to each other add to 25. Find them by guess, check and improve.

- 43 Write  $7mn$  in expanded form.

from 5.03

- 44 A number multiplied by itself, plus the number, is 90. Find the number.

**Challenging**

Questions 45–54

- 45 Two numbers that are next to each other add to 21. Find them by guess, check and improve.

- 46 A number multiplied by itself, plus the number, is 20. Find the number.

- 47  $3 \times \frac{2}{3}$

from 4.06

- 48 Guess, check and improve:  $3a + 18 = 57$ . Record each guess.

- 49 Write an expression: 4 less than  $c$

from 5.04

- 50 Solve by guess and check:  $n^2 = 144$ , where  $n$  is positive.

- 51 Two numbers that are next to each other add to 15. Find them by guess, check and improve.

- 52 Are  $6a^2$  and  $6a$  like terms?

from 5.07

- 53 A number multiplied by itself, plus the number, is 12. Find the number.

- 54 Find  $ab$  when  $a = 9$  and  $b = 6$ .

from 5.05

# 5.11 Solving one-step equations

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NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

To solve an equation, do the opposite operation to both sides: + undoes –, – undoes +, ÷ undoes ×, × undoes ÷. Check by substituting the answer.

Watch your teacher.  
Copy every step.

E1 Solve  $x + 9 = 23$ .

E2 Solve  $4m = 36$ .

E3 Solve  $\frac{p}{5} = 7$ .

## Easy

Questions 1–9

1 Solve  $t + 2 = 21$ .

2 Solve  $n - 20 = 11$ .

3 What undoes  $\div 5$ ?

4 Solve  $k + 8 = 30$ .

5 Solve  $n - 15 = 9$ .

6 What undoes  $+ 7$ ?

7 Solve  $y + 2 = 23$ .

8 Solve  $x - 17 = 15$ .

9 Solve in your head:  $\frac{t}{5} = 6$

from 5.10

## Easy

Questions 10–18

10 Solve  $p - 20 = 29$ .

11 What undoes  $- 4$ ?

12 Solve  $a + 5 = 17$ .

13 Evaluate  $7^2$ .

from 3.07

14 Solve  $3m = 12$ .

$3m \div 3 = \underline{\hspace{2cm}}$

$m = \underline{\hspace{2cm}}$

15 Solve  $a - 1 = 17$ .

16 What undoes  $\times 3$ ?

17  $kx \div k$

from 5.09

18 Solve  $t + 18 = 24$ .

## Medium

Questions 19–26

19 Solve  $4m = 36$ .

20 Solve  $\frac{x}{8} = 8$ .

21  $35 + 47 + 65 + 40$

from 5.01

22 Solve  $y + 4 = 3$ .

23 Solve  $8a = 24$ .

24 A rectangle is  $5b$  cm long and  $3n$  cm wide. Write its area.

from 5.08

25 Solve  $\frac{x}{4} = 12$ .

26 Solve  $w + 12 = 8$ .

## Medium

Questions 27–37

27 Solve  $\frac{w}{6} = 2$ .

28 Solve  $t + 10 = 5$ .

29  $\frac{6}{9} - \frac{1}{9}$

from 4.03

30 Solve  $8k = 24$ .

31 Work out  $5 \times (50 + 3)$  two ways.

from 5.02

32 Solve  $2m = 23$ . Give the answer as a fraction or mixed numeral.

33 Solve  $\frac{a}{5} = 8$ .

34 Is  $w = 3$  a solution of  $3w + 6 = 12$ ?

from 5.10

35 Solve  $n + 5 = 0$ .

36 Simplify  $\frac{16ay}{4y}$ .

from 5.09

37 Solve  $7k = 77$ .

## Challenging

Questions 38–47

38 Solve  $9p = 25$ . Give the answer as a fraction or mixed numeral.

39 Solve  $3a = -9$ .

40  $202 \times 6$

from 3.02

41 Solve  $56 = k + 47$ .

42  $4c \times c$

from 5.08

43 Ruby solved  $6a = 12$  and got  $a = 6$ . What went wrong? What is the answer?

44 Solve  $2n = 29$ . Give the answer as a fraction or mixed numeral.

45 Write  $9nx$  in expanded form.

from 5.03

46 Solve  $8w = -32$ .

47 Is  $m = 9$  a solution of  $4m + 10 = 46$ ? Show the check.

from 5.10

## Challenging

Questions 48–57

48 Solve  $5t = -15$ .

49 Solve  $34 = t + 26$ .

50  $50\,000 \div 1\,000$

from 3.03

51 Lena solved  $2t = 6$  and got  $t = 4$ . What went wrong? What is the answer?

52 Write an expression: 3 less than  $c$

from 5.04

53 Solve  $4y = 11$ . Give the answer as a fraction or mixed numeral.

54 Solve  $w + 11 = 4$ .

55  $16m \div 4$

from 5.09

56 Solve  $8p = -40$ .

57  $5mx \times 3m$

from 5.08

# 5.12 Solving two-step equations

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NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

A two-step equation: undo the + or – first, then the  $\times$  or  $\div$ . Do the same to both sides each time, then check your answer.

Watch your teacher.  
Copy every step.

**E1** Solve  $3x + 5 = 26$ .

**E2** Solve  $5m - 4 = 31$ .

**E3** Solve  $2(a + 3) = 18$ .

## Easy

Questions 1–9

**1** What do you undo first in  $5w + 9 = 34$ ?

**2** Solve  $3m + 7 = 16$ .

**3** Check: is  $p = 7$  the solution of  $5p - 10 = 20$ ?

**4** What do you undo first in  $8a - 3 = 27$ ?

**5** Solve  $5y + 7 = 32$ .

**6** Check: is  $y = 9$  the solution of  $6y - 6 = 42$ ?

**7** What do you undo first in  $4y - 8 = 31$ ?

**8** Solve  $5y + 3 = 8$ .

**9** Solve  $p - 10 = 17$ .

from 5.11

## Easy

Questions 10–18

**10** Solve  $5a + 7 = 17$ .

**11** Check: is  $t = 5$  the solution of  $6t - 12 = 12$ ?

**12** What do you undo first in  $6p + 13 = 58$ ?

**13** Write 5 out of 18 as a fraction in simplest form.

from 4.12

**14** Solve  $9m - 20 = 43$ .

$9m = \underline{\hspace{2cm}}$

$m = \underline{\hspace{1cm}} \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

**15** Solve  $2a + 5 = 11$ .

**16** Check: is  $b = 2$  the solution of  $2b - 5 = -1$ ?

**17** Solve in your head:  $\frac{x}{4} = 4$

from 5.10

**18** What do you undo first in  $4m + 9 = 42$ ?

## Medium

Questions 19–26

**19** Solve  $8x + 15 = 39$ .

**20** Solve  $\frac{t}{2} + 5 = 11$ .

**21**  $18 + 83 + 82 + 23$

from 5.01

**22** Solve  $2(n + 8) = 20$ .

**23** Solve  $5y + 16 = 46$ .

**24** Simplify  $\frac{20n}{5}$ .

from 5.09

**25** Solve  $\frac{m}{4} + 6 = 10$ .

**26** Solve  $4(p + 3) = 48$ .

## Medium

Questions 27–37

27 Solve  $\frac{a}{2} + 3 = 6$ .

28 Solve  $6(y + 7) = 96$ .

29  $7 \times \frac{1}{9}$

from 4.06

30 Solve  $7m - 18 = 59$ .

31 Work out  $9 \times (20 + 8)$  two ways.

from 5.02

32 Solve  $3a + 10 = 1$ .

33 Solve  $\frac{m}{6} + 5 = 13$ .

34 What undoes  $-4$ ?

from 5.11

35 Solve  $4(n + 8) = 60$ .

36 Solve by guess and check:  
 $a^2 = 49$ , where  $a$  is positive.

from 5.10

37 Solve  $8n - 14 = 18$ .

## Challenging

Questions 38–47

38 Solve  $6x + 7 = -29$ .

39 Solve  $9x + 12 = 24$ . Give the answer as a mixed numeral.

40  $412 \times 4$

from 3.02

41 Solve  $2(t - 4) = 6$ .

42  $4b \div b$

from 5.09

43 Mia solved  $6t + 5 = 23$  and got  $t = 12$ . What went wrong?

44 Solve  $6b + 4 = -50$ .

45 Write  $3cx$  in expanded form.

from 5.03

46 Solve  $8b + 11 = 25$ . Give the answer as a mixed numeral.

47 Solve  $3a = 33$ .

from 5.11

## Challenging

Questions 48–57

48 Solve  $3m + 15 = 22$ . Give the answer as a mixed numeral.

49 Solve  $3(x - 7) = -15$ .

50 What is 10% of 358?

from 4.11

51 Priya solved  $4p + 7 = 27$  and got  $p = 16$ . What went wrong?

52 Write an expression: 10 less than  $c$

from 5.04

53 Solve  $3y + 11 = 5$ .

54 Solve  $5(k + 9) = 105$ .

55 Solve in your head:  $y - 4 = 5$

from 5.10

56 Solve  $8t + 5 = 49$ . Give the answer as a mixed numeral.

57  $54ct \div 6t$

from 5.09

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

Watch your teacher.  
Copy every step.

## KEY IDEA

To solve a problem with an equation: choose a pronumeral for the unknown, write an equation from the words, solve it, then answer in a sentence.

**E1** A number times 4, plus 7, is 43. Find the number.

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**E2** 6 equal packs hold 72 pencils. How many pencils are in each pack?

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**E3** Two angles of a triangle are  $50^\circ$  and  $65^\circ$ . Find the third angle.

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## Easy

Questions 1–9

**1** Write an equation: A number plus 4 is 19.

**2** A number plus 2 is 14. What is the number?

**3** Write an equation: 5 times a number is 20.

**4** 7 times a number is 77. What is the number?

**5** Write an equation: A number minus 3 is 13.

**6** A number minus 4 is 5. What is the number?

**7** Write an equation: A number divided by 8 is 7.

**8** A number plus 4 is 21. What is the number?

**9** Solve  $2k + 4 = 12$ .

from 5.12

## Easy

Questions 10–18

**10** 4 times a number is 68. What is the number?

**11** Write an equation: A number plus 4 is 12.

**12** Double a number, plus 6, is 44. Write an equation and find the number.

$$\_\_\_\_\_\_ n + \_\_\_\_\_\_ = \_\_\_\_\_\_$$

$$n = \_\_\_\_\_\_$$

**13** Write  $3\frac{1}{2}$  as an improper fraction.

from 4.01

**14** A number minus 3 is 12. What is the number?

**15** Write an equation: 7 times a number is 112.

**16** Three times a number, minus 4, is 50. Write an equation and find the number.

**17** Write an expression: 7 less than  $b$

from 5.04

**18** A number plus 2 is 22. What is the number?

## Medium

Questions 19–24

**19** A number is multiplied by 5, then 9 is added. The answer is 59. Write an equation and find the number.

**20** 7 equal packs hold 203 pencils. How many pencils are in each pack?

**21**  $22 + 46 + 78 + 11$

from 5.01

**22** Two angles of a triangle are  $25^\circ$  and  $51^\circ$ . Write an equation and find the third angle.

**23** Double a number, plus 3, is 37. Write an equation and find the number.

**24** Solve  $\frac{k}{8} = 7$ .

from 5.11

## Medium

Questions 25–32

- 25 7 equal packs hold 56 pencils. How many pencils are in each pack?

- 26 Two angles of a triangle are  $68^\circ$  and  $70^\circ$ . Write an equation and find the third angle.

- 27 A class has 12 girls and 15 boys. What fraction of the class are girls?

from 4.08

- 28 Three times a number, minus 5, is 31. Write an equation and find the number.

- 29 Work out  $5 \times (50 + 3)$  two ways.

from 5.02

- 30 A plumber charges \$25 to call, plus \$25 an hour. The bill is \$125. How many hours did the plumber work?

- 31 3 equal packs hold 87 pencils. How many pencils are in each pack?

- 32 Solve in your head:  $\frac{t}{2} = 9$

from 5.10

## Challenging

Questions 33–40

- 33 A plumber charges \$30 to call, plus \$49 an hour. The bill is \$275. How many hours did the plumber work?

- 34 A rectangle is 2 cm longer than it is wide. Its perimeter is 64 cm. Find its width.

- 35 Write 0.8 as a percentage and as a fraction.

from 4.10

- 36 Aisha is 6 years older than Sam. Their ages add to 18. How old is Sam?

- 37 Write an expression: 6 times  $a$

from 5.04

- 38 Three numbers in a row add to 39. Find them.

- 39 A plumber charges \$25 to call, plus \$25 an hour. The bill is \$75. How many hours did the plumber work?

- 40 Write  $4np$  in expanded form.

from 5.03

## Challenging

Questions 41–49

- 41 A rectangle is 8 cm longer than it is wide. Its perimeter is 56 cm. Find its width.

- 42 Ali is 6 years older than Sam. Their ages add to 20. How old is Sam?

- 43 Write 35 in Roman numerals.

from 3.06

- 44 Three numbers in a row add to 36. Find them.

- 45 Find  $m^2$  when  $m = 5$ .

from 5.05

- 46 A plumber charges \$40 to call, plus \$48 an hour. The bill is \$136. How many hours did the plumber work?

- 47 Two angles of a triangle are  $47^\circ$  and  $20^\circ$ . Write an equation and find the third angle.

- 48 Is  $b = 7$  a solution of  $5b + 10 = 50$ ?

from 5.10

- 49 A rectangle is 7 cm longer than it is wide. Its perimeter is 54 cm. Find its width.